SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA15 – CLOUD COMPUTING AND BIG DATA ANALYTICS

ASSIGNMENT 1 – CO1

Design, Analyse and Validate a Virtualized Cloud Infrastructure for Efficient Resource Allocation and Data Centre Management.

CO1: Apply cloud computing technologies including hardware–software evolution, virtualization, web services, and service models to develop cloud-based solutions.

Field	Details	Field	Details
Student Name		Register Number	
Department / Section		Academic Year / Semester	
Faculty Name	Dr.Arul Raj A.M	Nature of Assignment	Individual
Duration	1 Week	Total Marks	50
Submission Date		Evaluation Date	

MARK SHEET (Rubric-based evaluation)

Task	Title	Max	Awarded	Evidence / evaluation focus
1	Existing Data Centre and Requirements	3		Definition; characteristics; deployment models; scenario mapping
2	Virtualization Fundamentals and Hypervisors	3		Timeline; hardware evolution; relevance to university
3	Server Consolidation and Resource Calculations	3		Web/software evolution; SOA; APIs; scenario relevance
4	Virtual Machines and Containers	5		Hypervisor types; VM/container comparison; consolidation; limitations
5	Virtualized Storage and Network Infrastructure	4		SOA and SOAP/REST; three university APIs; request/response examples
6	Resource Allocation, Scheduling and Monitoring	5		Responsibility layers; five-application mapping; comparison; providers
7	Availability, Security and Disaster Recovery	3		Four or more variants; two justified choices; provider risks
8	Virtualized Cloud Data Centre Architecture	5		Labelled diagram; deployment/service models; scaling; security; DR
9	Hands-on Implementation and Validation	4		Running VM/container; PaaS/SaaS; tests; actual screenshots
10	Cost, Performance and	7		Three-way comparison; 3-year costs;

	Final Recommendation			risks; phased roadmap
11	Day-4 Design Challenge	8		Impact; evaluation; adaptation; evidence-based defence
	TOTAL	**50**		**Percentage :**

EVIDENCE VERIFICATION CHECKLIST

Tasks	**Evidence expected**	**Status (✓ / P / ✗)**	**Evaluator note**
1–3	Conceptual explanations, comparisons and relevant illustrations		Explanations and relevant scenario mapping documented
4	Virtualization calculations with stated assumptions and peak-load caution		Calculations present; practical capacity validation recommended
5–7	API examples, service responsibility table and justified XaaS choices		Required conceptual evidence documented
8	Labelled architecture diagram, caption below figure and component justification		Architecture documented; sizing assumptions need testing
9	Actual code, environment details, execution logs, API outputs and screenshots		Procedures provided; actual execution evidence not established
10	Cost assumptions, 3-year totals, risk controls and migration roadmap		Estimates and rationale documented
11	Day-4 changes, 10× load strategy, 99.9% target and 40-college isolation		Design analysis documented; practical validation pending

IMPLEMENTATION EVIDENCE CHAIN

Evidence item	Status	**Observation**
Source code		Flask REST API and Dockerfile provided in the proposed implementation
Virtualization		Docker container configuration documented; running-container evidence required
PaaS or SaaS		Managed web-service deployment procedure described; live deployment not verified
Functional test		Expected HTTP 200 response and JSON output documented; actual output required
Failure case		HTTP 404 test case documented; actual execution record required
Screenshots		Screenshot requirements identified; authentic screenshots need verification
Reproducibility		Build commands, run commands, endpoints and expected results specified

MANDATORY SUBMISSION EVIDENCE CHECK

✓ = Present in documented answer; P = Partly present; ✗ = Missing / not satisfied

Task	Evidence expected	Status	Evaluator note
1	Cloud definition, characteristics, deployment models and scenario mapping		All conceptual requirements covered
2	Evolution timeline, hardware development and university relevance		Well explained with relevant examples
3	Internet software evolution, SOA and APIs		Clear explanation and scenario relevance
4	Hypervisors, VM/container comparison and consolidation calculations		Calculations included; peak-load limitations discussed
5	SOAP/REST comparison and three university APIs		API methods and request/response examples documented
6	IaaS/PaaS/SaaS responsibility layers and application mapping		All five applications categorized with justification
7	XaaS variants and two justified selections		Six variants explained; DBaaS and DRaaS recommended
8	Labelled architecture, scaling, security and disaster recovery		Design and component justifications documented
9	Working container, PaaS/SaaS deployment, actual tests and screenshots		Code and procedures provided; actual execution evidence requires verification
10	Three-year cost comparison, risks and migration roadmap		Cost assumptions, calculations and recommendations included
11	Impact analysis, 10× load, 99.9% availability and tenant isolation		Changed requirements evaluated; load-test evidence recommended

IMPLEMENTATION EVIDENCE CHAIN

Evidence item	Status	Observation
Source code		Python Flask REST API and Dockerfile documented
Virtualization		Docker build and run procedures provided; execution screenshot required
PaaS or SaaS		Cloud deployment steps described; actual deployment confirmation required
Functional test		HTTP 200 and expected JSON response documented; actual test output required
Failure case		HTTP 404 test case included; execution evidence required
Screenshots		Required screenshots identified but authenticity and completeness not verified
Reproducibility		Installation, configuration, commands and testing steps documented
Interpretation		Technical outcomes and implementation limitations explained
Code and report consistency		Proposed API, Docker configuration and testing procedure are aligned

SUBMISSION CHECKLIST (Not Separately Marked)

Requirement	Status
All 11 tasks answered	☐ Yes ☐ Partly ☐ No
☐ Existing data-centre requirements and limitations are identified.	☐Yes ☐ Partly ☐ No
☐ Virtualization and hypervisor types are explained.	☐Yes ☐ Partly ☐ No
☐ Server consolidation calculations include working and assumptions.	☐Yes ☐ Partly ☐ No
☐ VM and container comparison and workload mapping are included.	☐Yes ☐ Partly ☐ No
☐ Storage and network virtualization are addressed.	☐ Yes ☐Partly ☐ No
☐ Resource allocation and monitoring policies are included.	☐Yes ☐ Partly ☐ No
☐ Security risks, backup and disaster recovery are discussed.	☐Yes ☐ Partly ☐ No
☐ A labelled virtualized data-centre architecture diagram is included.	☐ Yes ☐Partly ☐ No
☐ Implementation screenshots and actual test evidence are provided.	☐ Yes ☐ Partly ☐ No
☐ Three-year cost analysis and migration roadmap are included.	☐ Yes ☐ Partly ☐ No

GENERAL INSTRUCTIONS

- Use clear headings and answer each task in the order given.
- Include diagrams, tables, calculations and screenshots wherever requested.
- State assumptions and units for all numerical calculations.
- Use credible technical references and cite sources consistently.
- Ensure that implementation evidence reflects work actually performed by the team.
- Submit the report as a well-formatted document with a title page, contents, references and appendices where required.

OVERALL REMARKS

_____ **Student Signature**	_____ **Faculty Signature**	_____ **Program Director**

CSA15 – Cloud Computing and Big Data Analytics

ASSIGNMENT 1

CO1: *Apply cloud computing technologies including hardware–software evolution, virtualization, web services, and service models to develop cloud-based solutions.*

Assignment Title:

Design, Analyse and Validate a Virtualized Cloud Infrastructure for Efficient Resource Allocation and Data Centre Management

(Total Marks – 50) Duration: 1 week

Scenario – University Data Centre Virtualization

A growing university operates an on-premises data centre that hosts its Student Information System, Learning Management System, Library Management System, Examination Portal, research applications and institutional website. Each major application runs on a separate physical server. Server resources are underutilized during normal periods, while admission, examination and result-publication periods create high demand. The university experiences rising electricity and maintenance costs, slow hardware procurement, limited scalability and inadequate disaster-recovery arrangements.

The current data-centre environment is assumed to have the following characteristics:

- 20 physical servers, each with 16 CPU cores, 64 GB RAM and 2 TB local storage.
- Average CPU utilisation is approximately 20%; peak utilisation reaches 90–95% during admission and examination periods.
- Most servers are dedicated to individual applications and cannot share unused capacity efficiently.
- There is no centralized virtual-machine (VM) management platform and no tested live-migration process.
- Backup is performed periodically, but a separate disaster-recovery site and documented recovery objectives are not available.
- The university wants to reduce infrastructure costs while improving availability, scalability, security and resource utilization.

You are appointed as a cloud infrastructure consultant. Analyse the existing environment and design a virtualized cloud data-centre solution. Your proposal must explain the role of hypervisors, virtual machines, containers, storage and networking virtualization, resource allocation, monitoring, security and disaster recovery. Justify all major decisions using calculations, diagrams and practical evidence.

Design Challenge:

On Day 4, the university announces that it will introduce a new digital learning platform and extend selected services to 10 affiliated colleges. The expected peak workload will increase to five times the current peak. The university requires at least 99.9% service availability, stronger tenant isolation and controlled operating costs. Re-evaluate your original design and decide whether it should be retained, modified or redesigned..

TASK 1 – Analyse the Existing Data Centre and Identify Requirements (3 Marks)

Study the current data-centre environment and identify the reasons for adopting virtualization.

Explain the purpose of a data centre and identify its major components: compute, storage, networking, power, cooling and management.

Identify at least four limitations of the existing physical-server environment.
Define resource utilization, server consolidation, scalability, availability and fault tolerance.
Prepare a requirement table covering performance, capacity, security, availability and cost.
Evidence Expected: A concise assessment of the existing environment, clearly linking each identified problem to a virtualization or cloud-infrastructure requirement.

TASK 2 – Explain Virtualization Fundamentals and Hypervisors (4 Marks)

Explain the technologies that enable multiple computing environments to share physical infrastructure.
Define virtualization and explain its purpose in a cloud data centre.
Explain the role and working of a hypervisor.
Differentiate Type-1 (bare-metal) and Type-2 (hosted) hypervisors with suitable examples such as VMware ESXi, Microsoft Hyper-V, KVM and Oracle VirtualBox.
Compare full virtualization, para-virtualization, hardware-assisted virtualization and OS-level virtualization (containers).
Recommend a suitable hypervisor for the university and justify the selection.
Evidence Expected: A labelled virtualization architecture diagram, comparison table and justified hypervisor recommendation.

TASK 3 – Design Server Consolidation and Calculate Resource Requirements (5 Marks)

Propose a consolidation plan to reduce the number of physical servers while maintaining sufficient capacity.
Explain physical-to-virtual (P2V) migration and server consolidation.
Estimate the total CPU and memory demand using the stated server specifications and average utilization. State all assumptions.
Assume a target average CPU utilization of 65% on the consolidated hosts.
Calculate the minimum number of hosts required based on CPU capacity, then check whether memory capacity changes the result.
Include a reasonable capacity reserve for peak demand and host failure; show the calculation clearly.
Discuss resource overcommitment, CPU scheduling, memory ballooning and the noisy-neighbour problem.
Evidence Expected: Worked calculations showing CPU and memory requirements, assumptions, capacity reserve and the recommended number of physical hosts. A final number without working will not receive full credit.

TASK 4 – Compare Virtual Machines and Containers (4 Marks)

Determine which workloads should run in virtual machines and which can run in containers.
Explain the architecture and characteristics of virtual machines and containers.
Compare VM and container isolation, operating-system requirements, startup time, resource overhead, portability and management.
Map at least five university workloads (for example, database, legacy student system, web portal, learning platform and analytics service) to VMs or containers.

Explain how Docker and a container orchestration platform such as Kubernetes can support deployment and scaling.
Identify situations where a VM is more appropriate than a container.
Evidence Expected: A VM-versus-container comparison and a workload mapping table supported by technical justification.

TASK 5 – Design Virtualized Storage and Network Infrastructure (4 Marks)

Design storage and network resources that support the proposed virtualized data centre.
Explain storage virtualization and compare direct-attached storage (DAS), network-attached storage (NAS) and storage area networks (SAN).
Explain software-defined storage (SDS), thin provisioning, snapshots, replication and backup.
Explain network virtualization, virtual switches, VLANs, virtual networks and software-defined networking (SDN).
Propose separate logical network segments for administration, student services, databases and management traffic.
Recommend suitable storage and network approaches for performance, security and availability.
Evidence Expected: A labelled storage and network design with explanations of the selected technologies and their roles.

TASK 6 – Plan Resource Allocation, Scheduling and Monitoring (5 Marks)

Develop a resource-management strategy that allocates infrastructure capacity efficiently.
Explain CPU, memory, storage and network resource allocation to VMs and containers.
Compare static allocation and dynamic allocation, including resource reservations, limits and shares.
Explain load balancing, workload scheduling, auto-scaling and high availability.
Identify suitable performance indicators: CPU utilization, memory utilization, disk I/O, network throughput, latency and VM availability.
Propose monitoring and alerting methods, and explain how the administrator should respond to resource contention.
Evidence Expected: A resource-allocation policy table and a monitoring plan that includes measurable indicators, thresholds and corrective actions.

TASK 7 – Evaluate Data Centre Availability, Security and Disaster Recovery (4 Marks)

Identify the risks associated with a virtualized data centre and propose suitable controls.
Explain hypervisor security, VM isolation, secure VM images, patch management and privileged access control.
Discuss VM escape, VM-to-VM attacks, insecure snapshots, VM sprawl and denial-of-service risks.
Recommend network segmentation, encryption, multi-factor authentication, role-based access control and audit logging.

Explain backup, replication, snapshots, live migration and disaster recovery.
Define Recovery Time Objective (RTO) and Recovery Point Objective (RPO), and propose suitable targets for at least two university applications.
Evidence Expected: A risk-and-mitigation table and a disaster-recovery plan with justified RTO and RPO values.

TASK 8 – Create the Virtualized Cloud Data Centre Architecture (5 Marks)

Construct an end-to-end architecture integrating compute, storage, network, virtualization and management components.
Draw a labelled architecture showing users, internet or campus network, firewall, load balancer, physical hosts, hypervisor layer, VMs/containers, virtual network, storage and monitoring.
Show how the university applications are distributed across hosts and how shared resources are allocated.
Include a management plane for provisioning, monitoring, logging and policy enforcement.
Show backup or replication to a secondary site or cloud region.
Justify the architecture in terms of scalability, performance, fault tolerance, security and cost.
Evidence Expected: A complete, clearly labelled architecture diagram and a written justification for each major component and design decision.

TASK 9 – Hands-on Implementation and Validation (5 Marks)

Build a small proof-of-concept that demonstrates virtualization and resource management.
Create at least one VM using VirtualBox, VMware, Hyper-V or KVM, or create a container using Docker.
Deploy a simple university-related service, such as a web page, student-record API or database-backed application.
Configure CPU, memory and storage resources and record the selected values.
Perform functional tests and at least one resource or load test. Record the observations.
Capture screenshots of the VM/container configuration, running service, resource-monitoring output and test results.
Document installation steps, commands, configuration details, issues faced and how they were resolved.
Evidence Expected: Reproducible implementation steps, source/configuration files, annotated screenshots and actual test evidence. Do not present simulated results as actual execution.

TASK 10 – Evaluate Cost, Performance and Final Recommendation (3 Marks)

Evaluate the proposed solution against the existing physical data-centre arrangement.
Compare the current physical-server model and proposed virtualized model in terms of hardware cost, electricity, cooling, maintenance, licensing and administration effort.

Prepare a simple three-year cost comparison using clearly stated assumptions.
Separate one-time capital expenditure from recurring operational expenditure.
Evaluate expected improvements in utilization, scalability, availability and manageability.
Identify risks such as licensing costs, vendor lock-in, skills requirements, hardware failure and migration downtime.
Provide a phased migration roadmap, identifying which applications should be migrated first and why.
Evidence Expected: A three-year cost table, risk analysis, migration roadmap and evidence-based final recommendation.

DAY-4 DESIGN CHALLENGE

TASK 11 – Re-evaluate and Adapt the Virtualized Infrastructure (8 Marks)

Changed Requirement

The university introduces a digital learning platform and extends selected services to 10 affiliated colleges. The expected peak workload increases to five times the current peak. The system must achieve 99.9% availability, provide secure isolation between colleges and remain within a limited operating budget.
Re-evaluate your original design. You may RETAIN, MODIFY or REDESIGN the proposed infrastructure.

Impact Analysis – 2 Marks

Identify changes in workload, users, storage, network traffic and availability requirements.
Explain how multi-tenant access affects VM/container isolation, network segmentation and access control.
Identify which original capacity and cost assumptions are no longer valid.

Capacity and Availability Evaluation – 2 Marks

Estimate the additional CPU, memory, storage and network capacity needed for the five-times peak workload.
Evaluate whether the proposed number of hosts and the existing failover arrangement are sufficient.
Explain the meaning of 99.9% availability (approximately 8.76 hours of downtime per year) and identify measures needed to approach it.

Adaptation or Redesign – 2 Marks

Decide whether to retain, modify or redesign the original infrastructure and justify the decision.
Describe the revised architecture. Consider clustering, live migration, auto-scaling, container orchestration, multi-zone deployment or hybrid-cloud capacity where appropriate.
Explain changes to resource-allocation policies, tenant isolation, backup and disaster recovery.

Evidence-Based Defence – 2 Marks

Support the revised design with capacity calculations and clearly stated assumptions.
Compare the cost and operational complexity of the original and revised designs.

Discuss trade-offs involving performance, availability, security, licensing and management effort.
Evidence Expected: A technically justified decision supported by revised architecture, capacity calculations, availability considerations, security controls and cost comparison.
.Re-evaluate your original design in response to these changed conditions. You may:

RETAIN → MODIFY → or REDESIGN

your original solution.

Impact Analysis – 2 Marks
Discuss:

- What has changed from the original problem
- Which original assumptions (load, users, availability, cost) are affected
- Why the web services and APIs now become critical
- How multi-tenant access by 40 colleges affects security and isolation

Evidence Expected: A clear analysis of how the changed requirements affect the original cloud design.

Evaluation of the Original Design – 2 Marks
Discuss:

- Whether the chosen service models and virtualization plan can handle a 10x load
- Whether auto-scaling and the single-region deployment are sufficient for 99.9% availability (about 8.76 hours of downtime allowed per year)
- Whether cost remains acceptable under the new load
- Whether another service model or XaaS offering would give better overall results

Evidence Expected: A technically supported evaluation of the original design under the new conditions.

Adaptation or Redesign – 2 Marks
Discuss:

- Decide whether to retain, modify or redesign the solution
- Describe the revised architecture (for example API gateway, container orchestration, serverless components, multi-zone deployment or a hybrid model)
- Explain what changed in the service model mix and in the virtualization approach
- If you retain the original design, give sufficient technical justification for why no change is required

Evidence Expected: An appropriate design response to the changed requirement, with clear technical reasoning.

Evidence-Based Defence – 2 Marks
Discuss:

- Capacity and availability calculations
- Cost comparison between the original and the revised design
- Security and multi-tenancy considerations

- Trade-offs such as vendor lock-in, complexity and migration effort

Evidence Expected: A well-supported final decision that follows logically from the analysis.

MARK DISTRIBUTION

Task	Title	Bloom's Level	Marks
Task 1	Existing Data Centre and Requirements	L2 – Understand / Analyse	3
Task 2	Virtualization Fundamentals and Hypervisors	L2 – Understand	3
Task 3	Server Consolidation and Resource Calculations	L3 – Apply	3
Task 4	Virtual Machines and Containers	L4 – Analyse	5
Task 5	Virtualized Storage and Network Infrastructure	L3/L4 – Apply / Analyse	4
Task 6	Resource Allocation, Scheduling and Monitoring	L4 – Analyse	5
Task 7	Availability, Security and Disaster Recovery	L4 – Analyse	3
Task 8	Virtualized Cloud Data Centre Architecture	L6 – Create	5
Task 9	Hands-on Implementation and Validation	L3 – Apply	4
Task 10	Cost, Performance and Final Recommendation	L5 – Evaluate	7
	Subtotal		

Task 11	Day-4 Design Challenge	L4/L5/L6 – Analyse / Evaluate / Create	8
	TOTAL		50

ANSWER

Task 1

1. Cloud Federation, Conventional Data Centre, SDDC and SOA

Introduction

Cloud federation is the interconnection of multiple independent cloud data centres to share resources and provide services. It helps an organization manage workloads dynamically across different geographical locations.

1. Conventional Cloud Data Centre Architecture

A conventional cloud data centre consists of physical servers, storage systems, networking devices and virtualization software.

- Physical servers provide CPU, memory and storage resources.
- Virtualization enables multiple virtual machines (VMs) to run on a single physical server.
- Load balancers distribute workloads among available servers.
- Resource managers allocate resources based on workload demand.
- Cloud federation allows resources from different data centres to be shared.

During festival sales, workloads can be moved from overloaded data centres to underutilized data centres.

2. Software-Defined Data Centre (SDDC)

SDDC virtualizes and manages computing, storage and networking resources through software.

- Software-defined compute dynamically allocates CPU and memory.
- Software-defined storage provides flexible storage allocation.
- Software-defined networking (SDN) manages network traffic.
- Centralized controllers automate resource allocation.
- Policy-based management supports workload migration and load balancing.

SDDC enables faster resource provisioning and better utilization across federated data centres.

3. Role of SOA

Service-Oriented Architecture (SOA) enables applications to communicate through independent, reusable services.

- Each data centre exposes services through standard APIs.
- REST or SOAP enables communication between services.
- Service registries help discover available services.
- Message queues support asynchronous communication.
- Loose coupling allows services to operate independently.

For example, an order-processing service in one data centre can communicate with an inventory service in another data centre.

4. Challenges and Solutions

Challenge	Solution
Resource sharing	Common resource management policies
Service availability	Replication, failover and load balancing
Identity privacy	Federated identity management
Security threats	Encryption, firewalls and zero-trust security
Network latency	Geographically distributed data centres
Scalability	Auto-scaling and workload migration

Conclusion

Cloud federation, SDDC and SOA together improve scalability, performance, resource utilization and service availability while maintaining security across multiple data centres.

Task-2

Given Data:

Number of physical servers = 15

CPU utilization per server = 25%

Power consumption per server = 350 W

New host power consumption = 450 W

Target utilization per host = 65%

Electricity cost = ₹9/kWh

Operating hours = 24 × 365 = 8760 hours/year

Assumptions:

All servers have equal CPU capacity. The new hosts have sufficient capacity to accommodate the workload. One additional host is maintained for N+1 fault tolerance. Power consumption is assumed to be constant.

(a) Number of Active Hosts and Total Hosts

Total workload:

15 × 25% = 375%

Number of active hosts:

N = 375 / 65 = 5.77

Rounding up,

Number of active hosts = 6

Including N+1 fault tolerance:

6 + 1 = 7 hosts

Answer: 6 active hosts and 7 total hosts.

(b) Consolidation Ratio

Consolidation Ratio = Old Servers / Active New Hosts

= 15 / 6

= 2.5:1

Answer: 2.5:1

(c) CPU Utilization

When all 7 hosts are active:

CPU Utilization = 375 / 7

= 53.57%

After one host fails, 6 hosts remain:

CPU Utilization = 375 / 6

= 62.5%

Answer:

Normal utilization = 53.57%

Utilization after one host fails = 62.5%

(d) Annual Electricity Cost Saving

Old system power:

15 × 350 = 5250 W

= 5.25 kW

New system power:

7 × 450 = 3150 W

= 3.15 kW

Annual cost of old system:

5.25 × 8760 × 9

= ₹4,13,910

Annual cost of new system:

3.15 × 8760 × 9

= ₹2,48,346

Annual saving:

₹4,13,910 − ₹2,48,346

= ₹1,65,564

Conclusion:

Virtualization reduces the number of physical servers and improves resource utilization. The company requires 7 hosts, including N+1 fault tolerance, and saves ₹1,65,564 annually in electricity costs.

Task-3

Given Data:

Physical CPU cores = 24

Physical RAM = 96 GB

Hypervisor reservation = 8 GB

Available RAM = 96 − 8 = 88 GB

Number of VMs = 12

vCPU per VM = 3

RAM per VM = 8 GB

(a) vCPU and Memory Over-Commitment Ratios

Total allocated vCPU:

12 × 3 = 36 vCPUs

CPU over-commitment ratio:

36 / 24 = 1.5

CPU over-commitment ratio = 1.5:1

Total allocated memory:

12 × 8 = 96 GB

Available memory:

96 − 8 = 88 GB

Memory over-commitment ratio:

96 / 88 = 1.09

Memory over-commitment ratio = 1.09:1

(b) Average and Peak Memory Sufficiency

Normal operation:

Each VM uses 5 GB RAM.

Total memory usage:

12 × 5 = 60 GB

Available RAM = 88 GB

Free memory:

88 − 60 = 28 GB

Therefore, normal memory operation is sufficient.

Peak operation:

Eight VMs use 8 GB each, and four VMs use 6 GB each.

Total memory usage:

(8 × 8) + (4 × 6)

= 64 + 24

= 88 GB

Available RAM = 88 GB

Free memory:

88 − 88 = 0 GB

Therefore, peak memory demand exactly matches available memory. No free memory remains.

(c) Average and Peak CPU Sufficiency

Normal CPU utilization:

Total allocated vCPU = 36

CPU usage = 35%

36 × 35% = 12.6 core-equivalents

CPU utilization:

(12.6 / 24) × 100

= 52.5%

Therefore, normal CPU operation is sufficient.

Peak CPU utilization:

36 × 75% = 27 core-equivalents

CPU utilization:

(27 / 24) × 100

= 112.5%

Peak CPU demand exceeds physical capacity by 3 core-equivalents.

Therefore, peak CPU demand cannot be fully satisfied simultaneously.

(d) Performance and Security Issues with Remedies

1. CPU contention can be reduced using CPU scheduling and workload balancing.
2. Memory exhaustion can be avoided by increasing RAM or reducing VM memory allocation.
3. Performance degradation can be prevented through continuous monitoring.
4. VM isolation can be improved using secure hypervisors.
5. Unauthorized access can be prevented using multi-factor authentication.
6. Data leakage can be prevented using encryption.
7. Resource exhaustion attacks can be controlled using quotas and monitoring.

Conclusion:

The host is sufficient during normal operation. Peak memory demand exactly matches available RAM, while peak CPU demand exceeds physical capacity. Additional resources and proper VM management are required.

Task-4

Introduction:

Cloud computing has developed through advances in computing hardware, virtualization, high-speed networks and Internet-based software.

Modern processors, large memory capacities and distributed storage systems enable efficient resource sharing. Virtualization allows multiple virtual machines to run on a single physical server. Internet software, web browsers and APIs enable users to access cloud services from anywhere.

1. Infrastructure as a Service (IaaS)

IaaS provides virtual machines, storage and networking resources.

StreamWave can migrate its existing content-management application to virtual machines while maintaining control over the operating system and installed software.

The cloud provider manages physical servers, storage and networking, while StreamWave manages the operating system and applications.

Justification: IaaS is suitable because StreamWave requires control over its operating system and installed software.

2. Platform as a Service (PaaS)

PaaS provides a platform for application development and deployment.

StreamWave can develop and deploy its new customer-facing application without maintaining servers or runtime environments.

The cloud provider manages servers, operating systems and runtime environments. Developers focus on application code.

Justification: PaaS is suitable because it reduces infrastructure management and simplifies application development.

3. Software as a Service (SaaS)

SaaS provides ready-made applications through the Internet.

StreamWave can subscribe to a CRM application and access it through a web browser. The provider manages the software, infrastructure, maintenance and updates.

Justification: SaaS is suitable because StreamWave wants to use an existing CRM application without developing or maintaining it.

4. XaaS – Everything as a Service

XaaS is a broad model in which different IT resources are provided as services.

Examples include:

- Storage as a Service (STaaS)
- Database as a Service (DBaaS)
- Network as a Service (NaaS)
- Security as a Service (SECaaS)
- Backup as a Service (BaaS)

XaaS enables StreamWave to obtain additional IT resources according to its business requirements.

Conclusion:

IaaS provides infrastructure control, PaaS simplifies application development, and SaaS provides ready-made applications. XaaS extends cloud computing by offering additional IT resources as services.

Task-5:

Given Data:

Number of cameras = 150

Clip size = 4 MB

Clip interval = 3 minutes

Storage cost = ₹1.50 per GB-month

Serverless cost = ₹0.00015 per invocation

Filtering = 25% of clips retained

Month = 30 days

1 GB = 1000 MB

(a) Daily Data Volume and Average Bandwidth Without Filtering

Clips generated per camera per day:

(24 × 60) / 3 = 480 clips

Total clips per day:

150 × 480 = 72,000 clips

Daily data volume:

72,000 × 4 = 288,000 MB

= 288 GB/day

Average bandwidth:

(288 × 1000 × 8) / 86400

= 26.67 Mbps

(b) Monthly Storage Cost Without Filtering

Monthly storage:

288 × 30 = 8640 GB

Storage cost:

8640 × ₹1.50

= ₹12,960

(c) Daily Data Volume, Bandwidth and Storage Cost After Filtering

Only 25% of clips are uploaded.

Daily data volume:

288 × 0.25 = 72 GB/day

Average bandwidth:

26.67 × 0.25 = 6.67 Mbps

Monthly storage:

72 × 30 = 2160 GB

Storage cost:

2160 × ₹1.50

= ₹3,240

Monthly storage saving:

₹12,960 − ₹3,240

= ₹9,720

(d) Monthly Serverless Invocation Cost After Filtering

Daily invocations:

72,000 × 0.25 = 18,000

Monthly invocations:

18,000 × 30 = 540,000

Monthly serverless cost:

540,000 × ₹0.00015

= ₹81

Role of REST API:

REST API enables communication between cameras and cloud object storage using HTTP methods such as POST and PUT. It supports standardized data uploading, authentication and access to cloud resources.

Why Object Storage Is Suitable:

Object storage is suitable for video clips because it provides scalable, durable and reliable storage. It supports metadata, lifecycle management and efficient storage of large video files.

Conclusion:

Edge filtering reduces uploaded data, bandwidth usage, storage cost and serverless invocations by 75%. Object storage and REST APIs provide an efficient solution for video monitoring.

Task-6:

Introduction:

Cloud federation connects independent cloud platforms to share resources and services. An educational institution can use federation to integrate its Learning Management System (LMS), digital library and Student Information System (SIS) across three geographical locations.

1. Cloud Federation

Cloud federation enables independent cloud platforms to share resources and services.

- LMS provides online learning materials.

- Digital library provides books and research resources.
- Student Information System manages academic records.
- Federation enables unified access to these services.
- Resources can be shared across geographical locations.

2. Role of SDDC

SDDC virtualizes and manages computing, storage and networking resources through software.

- Software-defined compute dynamically allocates CPU and memory.
- Software-defined storage supports shared academic data.
- Software-defined networking connects services across cloud locations.
- Automation manages resource allocation.
- Load balancing improves performance.

3. Role of SOA

SOA enables different applications to communicate through reusable services.

For example, the LMS requests student details from the Student Information System. The digital library verifies user permissions through an identity service.

REST, SOAP and API gateways enable communication between services.

4. Challenges and Solutions

Federated identity: Use SAML, OAuth 2.0 and OpenID Connect.
User presence privacy: Minimize collected information and obtain user consent.
Data location: Use approved geographical storage locations.
Data privacy: Apply encryption and access control.
Service failure: Use replication and disaster recovery.
Network latency: Use regional cloud deployment.
5. Secure Integration

- Use single sign-on (SSO).
- Apply role-based access control (RBAC).
- Encrypt data during transmission and storage.
- Use multi-factor authentication.
- Maintain audit logs.
- Follow data protection policies.

Conclusion:
Cloud federation, SDDC and SOA provide unified educational services across geographical locations. Secure identity management, encryption and reliable infrastructure ensure efficient service integration.

Task-8:

Introduction:

Two independent universities can establish cloud federation to share research-computing resources and digital-library services while maintaining independent administration.

1. Conventional Cloud Data-Centre Architecture

A conventional cloud data centre includes physical servers, storage, networking devices and virtualization platforms.

- Virtual machines provide isolated environments.
- Resource managers allocate computing resources.
- Load balancers distribute workloads.
- Federation connects the two universities.
- Research workloads can be transferred to available resources.

2. Software-Defined Data Centre (SDDC)

SDDC provides software-based management of computing, storage and networking.

- Compute virtualization enables dynamic resource allocation.
- Storage virtualization supports shared research data.
- Software-defined networking establishes secure communication.
- Automated management improves resource utilization.
- Policy-based allocation supports research workloads.

3. Role of SOA

SOA enables independent services to communicate through standard interfaces.

- Research-computing services are exposed through APIs.
- Digital-library services are shared between universities.

- Service registries support service discovery.
- REST and SOAP enable communication.
- Loose coupling allows independent service management.

4. Federation Challenges and Solutions

Resource compatibility: Use common standards and APIs.

Federated identity: Use SAML-based identity federation.

Service availability: Use replication and failover.

Data privacy: Use encryption and access restrictions.

Resource sharing: Establish federation agreements and quotas.

Security threats: Use firewalls and zero-trust access.

5. Privacy Risks

Federated identity risks include unauthorized access, identity theft and excessive privilege assignment.

User presence information may reveal login times, locations, research activities and user behavior.

6. Technical and Administrative Safeguards

Technical safeguards include multi-factor authentication, role-based access control, encryption, secure APIs, audit logs and intrusion detection.

Administrative safeguards include data-sharing agreements, access policies, regular security audits, privacy policies, user consent and incident response procedures.

Conclusion:

Cloud federation, SDDC and SOA enable universities to share resources and services efficiently. Strong identity management, privacy protection and administrative policies ensure secure collaboration.

Task-8

Introduction:

Three healthcare organizations can establish a federated cloud environment to share medical research applications and diagnostic resources. Cloud federation enables independent data centres to collaborate while maintaining control over their resources.

1. Cloud Data-Centre Architecture

A cloud data centre consists of servers, storage, networks and virtualization software.

- Virtual machines host healthcare applications.
- Storage systems maintain medical research data.
- Resource managers allocate computing resources.
- Load balancers distribute workloads.
- Federation connects the three healthcare organizations.

This allows diagnostic workloads to be processed using available computing resources across organizations.

2. Role of SDDC

SDDC provides software-based management of infrastructure resources.

- Compute virtualization allocates CPU and memory dynamically.

- Software-defined storage enables scalable medical data storage.
- Software-defined networking provides secure communication.
- Automated resource management improves utilization.
- Network segmentation isolates sensitive healthcare workloads.

3. Role of SOA

SOA enables healthcare applications to communicate through reusable services.

- Diagnostic applications share medical research results.
- Electronic Health Record systems communicate through APIs.
- Research services access approved medical datasets.
- REST and SOAP enable communication between applications.
- Service registries support service discovery.

Healthcare systems can use standards such as HL7 and FHIR for exchanging medical information.

4. Security Challenges

Federated access: Unauthorized users may access medical services.

Identity management: Identity theft and impersonation may occur.

Patient data privacy: Sensitive medical records may be exposed.
Data sharing: Unauthorized disclosure of medical data may occur.
Network security: Communication between data centres may be attacked.
Availability: Service failures may affect diagnostic applications.

5. Authentication Mechanisms

- Multi-factor authentication (MFA).
- Federated identity management.
- Single sign-on (SSO).
- OAuth 2.0 and OpenID Connect.
- Mutual TLS for secure service communication.

6. Authorization Mechanisms

- Role-based access control (RBAC).
- Attribute-based access control (ABAC).
- Least-privilege access.
- Patient consent management.
- Fine-grained access policies.

7. Data Protection Mechanisms

- AES-256 encryption for data at rest.
- TLS encryption for data in transit.
- Data anonymization and pseudonymization.
- Secure key management.
- Audit logs and continuous monitoring.
- Backup and disaster recovery.

8. Administrative Safeguards

- Establish healthcare data-sharing agreements.
- Define patient consent and access policies.
- Conduct regular security audits.
- Train healthcare staff on data privacy.
- Follow applicable healthcare and data-protection regulations.

Conclusion:

Cloud federation, SDDC and SOA enable healthcare organizations to share diagnostic resources and medical research applications. Federated identity management, strong authentication, fine-grained authorization and encryption ensure secure and reliable healthcare service integration.